

Problem-Solution Fit

Date	15 March 2026
Team ID	XXXXXX
Project Name	AI Learning Support Assistant
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - College Students - University Students - Self-Learners - Online Course Learners - Programming Beginners - Students preparing for exams	4. EMOTIONS BEFORE ☹️ Confused ☹️ Frustrated ☹️ Anxious ☹️ Uncertain AFTER [EM] 😊 Confident 📖 Better informed 🎯 Focused 😊 Motivated	7. BEHAVIOR + ITS INTENSITY [BE] - Searches online for explanations (High) - Uses AI tools for learning (High) - Revisits previous study material (Medium) - Practices repeatedly to understand concepts (High)
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty understanding complex academic concepts (High) - Confusion while learning new topics (High)	5.AVAILABLE SOLUTIONS PROS & CONS [AS] Pros - Free learning resources - Wide range of educational content	8. CHANNELS OF BEHAVIOR [CH] Online - AI Learning Support Assistant - Google Search

- Lack of personalized learning guidance (High) - Difficulty tracking previous learning sessions (Medium) Unable to monitor learning progress (Medium)	Cons - Generic explanations - No emotion-aware support - No learning history	- YouTube - ChatGPT - Educational websites Offline - Classroom lectures - Teachers - Study groups - Books - Personal notes
---	---	--

3. TRIGGERS TO ACT [TR] - Unable to understand a topic - Difficulty solving assignments or practice problems - Preparing for examinations - Need for quick academic guidance - Desire to review previous learning sessions	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited study time - Different learning speeds - No immediate access to teachers - Limited computing resources Dependence on internet for AI responses	9. PROBLEM ROOT / CAUSE [RC] - Difficult academic concepts - Generic learning resources - Lack of personalized academic guidance - No emotion-aware learning assistance - Difficulty reviewing previous learning interactions

10. YOUR SOLUTION [SL]

The **AI Learning Support Assistant** helps students by providing personalized academic guidance based on their learning problems and detected emotions.

Implemented Features :

- Emotion prediction using **BiLSTM**
- Emotion prediction using **BERT**
- Mixed emotion detection

- Personalized learning guidance using **Google Gemini API**
- Emotion confidence analysis
- Interactive BiLSTM probability charts
- Previous interaction history
- Download interaction history (CSV)
- Analytics dashboard with Plotly visualizations
- Emotion distribution analysis
- Confidence trend analysis
- Subject-wise analytics
- Responsive Streamlit interface

Value Delivered :

- Personalized learning assistance
- Faster understanding of difficult concepts
- Emotion-aware academic support
- Learning history for revision
- Visual analytics for monitoring learning progress
- Simple and interactive user experience